Healthcare Companion Platform

Proposed scope for the first release

SOFTENG 761 · Team Agile Avengers · Prepared 16 August 2026 · Draft for your review

> Following our meeting on 11 August, this document sets out what we propose to build first, what we intend to leave until later, and the decisions we need from you before we start. Please treat every part of it as open to change — if we have misunderstood something, this is the cheapest moment to correct it.

1. What we understood from our meeting

Your group already runs a healthcare robot system, but its server and web pages sit on a university-managed machine inside the university network. The people who actually need it — patients, caregivers, and hospital medication specialists — cannot reach it, and it can no longer be maintained because the original developer has left. We are therefore building a new system from scratch, deployed somewhere your group controls and reachable from outside the university network.

Alongside that, we understood the following:

- The **web page is the first priority**, and it serves people — healthcare professionals entering medications and exercises, caregivers monitoring and helping the patient, and an administrator.
- For this project the robot is replaced by a **virtual agent** running on a phone or local machine. Your group can substitute a real robot later.
- **One device cares for exactly one patient.**
- The agent reminds the patient, checks their identity, and records what they report. It does not administer anything.
- The system is **trust-based by design**. It does not verify that medication was physically taken; the patient confirms, and a human caregiver provides the real check. We understand this was a deliberate choice, and we are not designing around it.
- Face recognition and other AI are **called from your CARE server** rather than built by us.
- A responsive web application that works on both phone and laptop is sufficient; native mobile apps are not required.

2. What the first release delivers

> A healthcare professional can set up a patient's medication and exercise plan in the web portal. The patient is reminded on their phone at the right time, identifies themselves, and confirms. The caregiver can see what was taken and what was missed — on a system reachable from outside the university network.

The journey it supports, end to end

1. A professional registers, signs in, and creates a patient record
2. They link a caregiver and assign one device to that patient
3. They enter a prescription — medications, doses, instructions, schedule — and regular exercises

4. The system generates that patient's daily schedule
5. At the scheduled time the device reminds the patient
6. The patient identifies themselves, with a second method available if the face check cannot be used
7. The patient confirms, skips, or postpones; the device reports this back to the server
8. The caregiver opens the portal and sees what was confirmed and what was missed

We have deliberately kept all four parts — portal, care plan, device, and oversight — in this first release. A portal with no device, or a device with no oversight, would not be usable by anyone. In particular, the caregiver's view of missed doses is what makes the trust-based model safe, so we do not think it can wait.

3. In the first release

Area	Included
Accounts and access	Registration and sign-in for professionals and caregivers; patient accounts created on their behalf; separate permissions per role; caregivers see only their own patients
Patients and caregivers	Create a patient record; link a caregiver to a patient; assign one device per patient
Care plan	Create prescriptions with medications, doses and instructions; set schedules; add regular exercises; amend or stop a plan; automatic generation of each patient's daily schedule
Web portal	Professional dashboard; combined patient view; simplified caregiver view; works on phone and laptop; consistent design and navigation; input validation
Patient device	Pairing to a patient; daily schedule; medication and exercise reminders; identity check with an alternative method; confirm, skip or postpone; reporting back to the server; a large, simple interface suitable for elderly users and children
Oversight	Full medication history with times; caregiver view of missed doses
Deployment	Reachable from outside the university network, over HTTPS, on infrastructure your group controls

4. Planned, but not in the first release

These are on our backlog and we intend to reach as many as we can. We have set them aside only to make the first release coherent and deliverable.

Feature	Reason for deferring
Live connection to the CARE server	We will build against a stand-in so that progress does not depend on CARE's completion date. Swapping in the real service is a small change once it is available.
CARE retrieving data from our system	We would like to agree the data format with Jay before building this, rather than guess it.
Photographs of medications shown to the patient	Valuable for exactly your users — the first thing we plan to add after the core is working.
Repeat reminders, spoken reminders, offline operation	Improvements to reminder reliability, once the delivery method is settled (see question 2).
Alerting a caregiver when a dose is missed	Depends on your answer to question 3.
Adherence reports over time, data export, interaction history	Deeper oversight, beyond "what was missed today".
Password reset, account deactivation, access audit log	Important for a production system; not required to demonstrate the product working.
Patient search, profile editing, visiting history, patient photographs	Breadth of record-keeping rather than core function.

> **One exclusion we want to flag explicitly.** Detailed accessibility work — keyboard operation, text scaling, colour contrast — is not listed as part of the minimum, but we do not regard it as optional. Your users are elderly people and children, so we are treating it as ongoing work throughout the project rather than a single task. We mention it here so it is visible rather than silently dropped.

5. What we are assuming

If any of these is wrong, please tell us — each one changes what we build.

1. The patient's device is a responsive web application, not a native app.
2. We may develop and demonstrate against a stand-in for the CARE server until it is ready.
3. Patient data used during development is entirely synthetic.
4. Exercises follow the same remind-and-confirm pattern as medications.
5. "One device per patient" is a firm rule we should enforce in the system's design.

6. What we need from you

1. Is this the right minimum?

In particular: should the caregiver's view be part of the first release, or would you rather we went deeper on the professional's side first?

2. How should a reminder actually reach the patient?

A reminder shown only inside the app appears solely while the patient already has it open — which does not help someone who has forgotten. The realistic options are an installable web app with push notifications, a native Android app, or email/SMS. Each carries different effort and limitations, and we would like to choose it with you.

3. When a dose is missed, should anyone be alerted?

Or is recording it for the caregiver to review sufficient? This determines how urgent the notification work is.

4. Will the system ever hold real patient data, or only synthetic data?

If real or anonymised data is involved at any stage, we need to know what ethics approval and privacy obligations apply to us before we design the data model.

5. Where will the system be hosted, and who provides it?

This is the one item that cannot start without you, and it is the whole point of the project. If a server or cloud account needs to be arranged, starting now matters — provisioning can be slow.

6. Is this project confidential?

We demonstrate our progress to the class each week. If any part of the project should not be shown publicly, please let us know and we will present it privately to the teaching team only.

7. Could you rank the four areas in order of importance?

Web portal, care plan, patient device, oversight. We have understood the portal to be first. Your ordering of the remaining three tells us what to protect if we run short of time.

7. Timeline and what we can commit to

The project runs as seven one-week sprints, which gives us approximately five to six weeks of actual development time with eight developers working alongside our other coursework. We will do our best to accommodate as many features as possible in that window, but we cannot guarantee a complete solution. If we finish the scope above early, we will come back and ask you for more.

Date	Milestone
24 August	First progress demonstration. We expect to show a working, deployed system with a real care plan in it.
14–21 September	Portal and patient device built out
5 October	Oversight features and refinement
Mid-October	Final demonstration and handover. You are very welcome to attend.

We would value a short conversation on the seven questions above at your earliest convenience, as questions 2, 4 and 5 affect work we would like to begin immediately. Anything you can answer by email in the meantime would help us keep moving.